PROJECT 04 / TRAVEL SUPPORT / IMPLEMENTATION WORKFLOW

OpsDesk

A travel-focused Zendesk replacement

A shared support desk that connects each new complaint to the customer's agency, booking, previous tickets and supplier discussions. Agents see the whole case before deciding what to do next.

1 RECEIVE
Save complaint

2 CONNECT
Booking + history

3 ROUTE
Team + agent

4 RESOLVE
Supplier + reply

CONTINUOUS: monitor deadlines, escalate, record events
Every saved event updates the case history and insights dashboard

The six business areas

Area	Required demo behavior
Connected history	Attach tickets to agency, customer and booking; show earlier contacts.
Sensible assignment	Route by region, issue type, availability and agent workload.
Supplier conversations	Maintain a separate supplier thread attached to the main ticket.
Travel deadlines	Track check-in, departure and relevant cancellation deadlines.
Escalation	Raise priority and notify a manager when a configured rule is triggered.
Insights	Show repeat complaints, supplier delays, backlog and breached deadlines.

Demo objective

Seed recurring refund cases, submit a related complaint, show automatic linking and routing, resolve a supplier conversation, trigger a deadline escalation, and explain the resulting dashboard trends.

Scope: a 3-5 day prototype. Rules below are proposed demo policies, not verified competition requirements or production service commitments.

01 / FOUNDATION

Reuse the helpdesk; build the travel logic

Resource	Recommended use
IT Helpdesk [1]	**Main codebase.** Reuse Next.js, NestJS, PostgreSQL/Prisma, authentication, roles, tickets, assignment, message threads and Gemini intake. Build the travel layer and analytics.
Chatwoot [2]	**Alternative foundation and UX reference.** Study shared inboxes, customer history, private notes, teams, assignment and reporting. Do not run it as a second ticket database.
Open HelpDesk [3]	**Workflow reference.** Study email, SLA monitoring, scheduled automations and reports. Its README identifies the project as alpha; enterprise features have separate licensing.
Bitext Travel [4]	Approximately 31,700 travel examples. Use selected complaint wording and intent labels for classification examples and evaluation.
Bitext Customer Support [5]	26,872 question/answer pairs. Use for general support-intent examples and a held-out evaluation set.

Recommended architecture

Layer	Responsibility
Next.js interface	Customer intake, agent inbox, ticket workspace, manager queue, insights.
NestJS backend	Authentication, agency isolation, linking, routing, message visibility and ticket lifecycle.
PostgreSQL	Durable bookings, tickets, messages, assignments, deadline records and event history.
Redis + background worker	Scheduled deadline checks, queued notifications and existing rate limits. Persist job outcomes in PostgreSQL.
Gemini API	Extend the existing intake integration for classification, history summaries and response drafts. Claude is an optional replacement.
Supplier channel	Start with an in-app supplier reply flow. Add bidirectional email once the core flow is reliable.

Repository decision

Use IT Helpdesk as the single application foundation. Reference the other two projects for designs and workflows instead of merging three full systems. Confirm the chosen repository revision, setup instructions and license before implementation.

02 / OPERATIONS

From incoming complaint to resolution

Step	System behavior
1. Receive and persist	Accept a customer/agency form or chat submission. Save the original complaint immediately, even if AI is unavailable.
2. Identify the booking	Match the booking reference within the authorized agency. Check the requester's access. If missing or ambiguous, ask for clarification or send to agent review.
3. Retrieve context	Load the booking, prior tickets, previous messages and authorized supplier updates. Display a timeline; keep separate complaints as linked tickets.
4. Classify and summarize	AI proposes the issue category and a short, source-linked summary. Store suggested fields separately from confirmed booking facts.
5. Prioritize and route	Rules select urgency, region/team and an eligible available agent with low workload. Otherwise retain the ticket in the unassigned queue.
6. Collaborate with supplier	Agent opens a dedicated supplier conversation. Save replies under that conversation, preserving internal-note and customer-message boundaries.
7. Monitor and escalate	A worker rechecks open cases against UTC deadlines. Record the triggering rule, notify the manager, and avoid repeat alerts for the same escalation.
8. Resolve and measure	Agent reviews and sends the response, records the resolution and closes the case. Update metrics from stored events; allow controlled reopening.

Ticket and conversation lifecycle

Main ticket: Open - In progress - Waiting for supplier/customer - Resolved - Closed. Reopening returns it to an active state and records a new event.

Supplier thread: Open - Awaiting reply - Replied - Closed. A supplier reply does not automatically resolve the customer's ticket.

Deadline handling

Store deadlines in UTC with the original booking timezone for display. Recalculate deadlines when travel details change. A refund after a completed trip follows its refund/SLA deadline rather than an unrelated departure date.

03 / RULES AND RECORDS

Make decisions explainable and access explicit

Proposed demo condition	Action
Unresolved service issue; check-in within 24 hours	Raise to High and show the deadline in the agent queue.
Unresolved service issue; check-in within 6 hours	Raise to Urgent, record escalation and notify the manager.
Supplier response target missed	Notify the owner; escalate according to the configured policy.
Repeated unresolved issue for the same booking	Flag recurrence and show related tickets; do not silently merge them.

Core records and relationships

Agency - Customer - Booking - Ticket forms the main relationship chain. Bookings also reference suppliers. Each ticket has customer messages, internal notes, supplier conversations, assignments, deadlines and events.

Record	Important fields
Booking	Agency/customer IDs, booking reference, supplier, region, travel/check-in timestamps, timezone.
Ticket	Agency, customer, booking, category, status, priority, team, agent, response deadline.
Message / supplier thread	Author, timestamp, content, visibility; parent ticket, supplier and response target.
Assignment / escalation	Previous/new owner or priority, triggering rule, reason, actor and timestamp.
Ticket event	Event type, ticket, actor, timestamp and before/after values for reporting and audit.

Access rules enforced by the backend

Role	Allowed scope
Customer	Own tickets and customer-visible messages only.
Agency user	Bookings and cases belonging to their authorized agency.
Agent	Authorized team queues and assigned cases, including permitted internal context.
Manager / admin	Authorized teams, reassignment, escalation, configuration and reporting.
Supplier	Only the supplier conversations explicitly shared with them; no customer history or internal notes.

Keep agency isolation on both direct record lookups and history searches. A matching booking-reference string alone must never grant access.

04 / INTELLIGENCE AND INTERFACE

Use AI for context; use rules for control

AI task	Expected output
Complaint classification	A validated category suggestion such as refund, cancellation or booking change.
History summary	A short summary referencing the relevant stored tickets and messages.
Related-ticket suggestion	Candidates for agent review when exact booking linking is unavailable.
Response drafting	An editable draft grounded in confirmed booking facts and supplier replies.

Example: "Third contact about booking BK-1042. Refund remains pending with Supplier A. The previous agent requested confirmation yesterday."

Permissions, verified booking links, deadline calculations and escalation decisions remain explicit backend rules. Agents review outgoing drafts. If the model fails, manual intake, routing and case handling must still work.

Dataset preparation

Use Bitext instructions as complaint examples and its intent/category labels for evaluation. Map the original labels to your application categories. Keep a separate held-out set; initial fine-tuning is unnecessary.

Bitext response text is not an authoritative supplier policy or proof of a real refund. Build synthetic relational data for bookings, repeat contacts, supplier replies, timestamps and assignment history.

Seed records	Suggested prototype size
Agencies / customers / bookings	3 agencies, 20 customers, 30 bookings across several regions.
Tickets / people / suppliers	60 historical and current tickets, 4 agents, 1 manager, 5 suppliers.
Scenario coverage	Repeat refund, missing booking reference, supplier delay, urgent check-in, resolved case, misleading identity match.

Five screens to implement

Agent inbox: Filters for region, priority, status, owner and deadline.

Ticket workspace: Customer thread, booking context, related history, summary and actions.

Supplier panel: Separate thread, reply status and response countdown.

Manager queue: Urgent, unassigned, overdue and escalated cases.

Insights: Repeat issues, backlog, response/resolution times, supplier delays and breaches.

05 / DELIVERY

Build plan, acceptance checks and sources

Day	Deliverable and completion check
1 / Foundation	Run IT Helpdesk. Add agencies, bookings and seed records. Verify persistence after restart and role/agency restrictions.
2 / Connected history	Implement booking linking, related-ticket timeline and summaries. A repeat complaint must reveal earlier context.
3 / Routing + supplier	Implement team/agent rules and separate supplier threads. Verify assignment and thread visibility.
4 / Deadlines	Add worker, escalation records and manager queue. Advance a demo clock; confirm one escalation without duplicate alerts.
5 / Insights + demo	Add event-based metrics, polish screens and run the full scenario. Verify dashboard totals against stored events.

Minimum acceptance checks

Tickets survive restart; agency A cannot access agency B; ambiguous references require review; supplier messages stay private to their intended participants; unavailable agents are skipped; deadline checks survive worker restart; escalation alerts are idempotent; AI failure permits manual work; dashboard numbers reconcile with stored records.

Three-day scope and final demo

For three days: use one region, in-app supplier replies and simple charts while retaining all six business areas. Simulate time through an explicit demo clock. Real email delivery, live booking integrations and model fine-tuning can follow later.

Demo: submit another complaint for BK-1042 - retrieve history - assign an agent - request supplier input - advance toward check-in - escalate - record resolution - show the trend on the dashboard.

Source links

[1] IT Helpdesk - codebase and current feature roadmap

[2] Chatwoot - collaboration, customer history and reporting

[3] Open HelpDesk - email, SLA and automation reference

[4] Bitext Travel - dataset card and downloadable files

[5] Bitext Customer Support - dataset card and files

Source review: 9 November 2026. Repository descriptions establish available starting features, not independent production-readiness validation. The workflow, seed sizes and escalation thresholds are proposed implementation choices.